

# Some useful definitions

**Sky localization map:** a file that contains a pixelated reconstruction of the joint distribution  $P(\text{RA}, \text{Dec})$ , plus a conditional posterior  $P(D_L | \text{RA}, \text{Dec})$  on the luminosity distance. With these 3 info, we can fully build the 3D localization of the source

**Sub Solar Mass search:** there are now 2 pipelines (GstLAL and MBTA) that perform an additional search on mass ranges where we expect exotic objects:

$$0.2M_{\odot} < m_2 < 1.0M_{\odot} \text{ and } 0.2M_{\odot} < m_1 < 10M_{\odot}$$

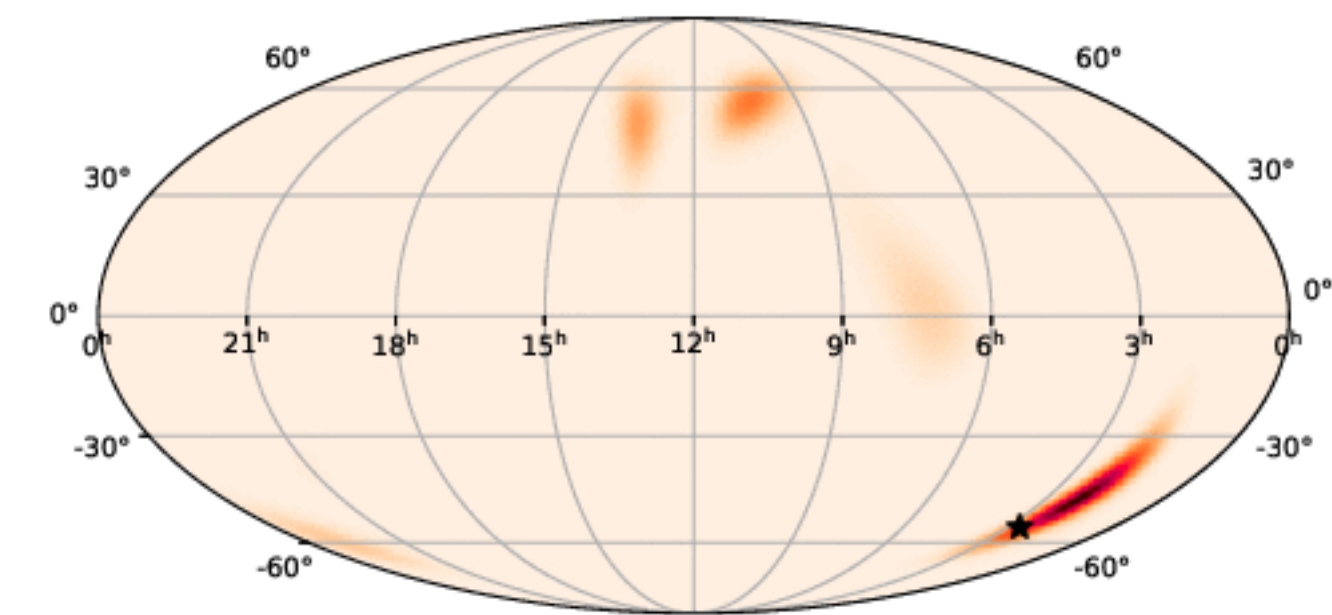
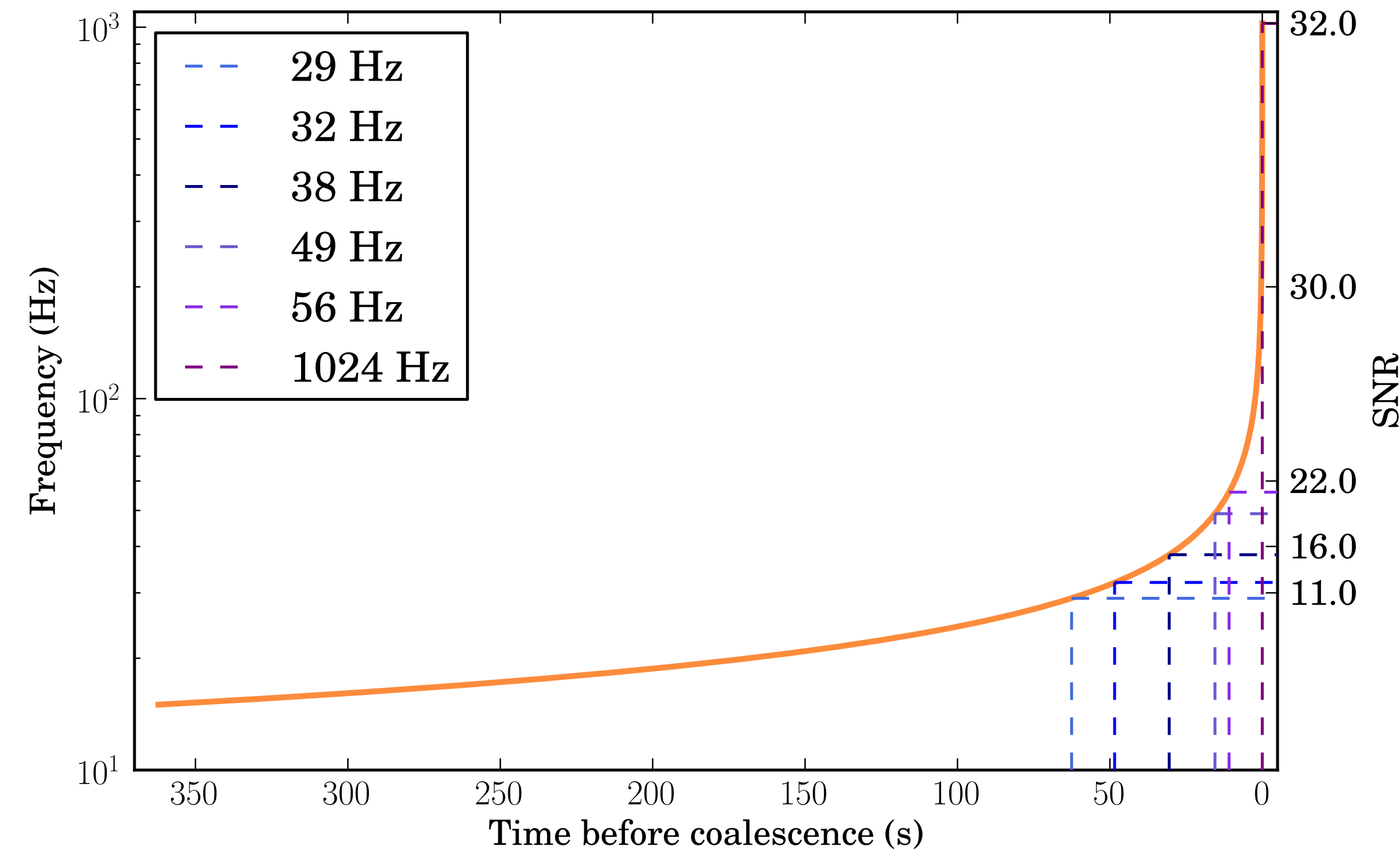
Caveat: an SSM search can also trigger on a BNS!!

**P(HasSSM)** is also now released as part of IGWN alerts, if an SSM search triggers, reporting the probability that at least one has mass  $< 1 M_{\text{sun}}$

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**Early Warning Alert:** For long and loud enough signals (less massive, longer inspiral), pipelines can accumulate enough SNR (above detection threshold) already before the merger

gw170817-like event



| Frequency | Localization accuracy (90% credible area) |
|-----------|-------------------------------------------|
| 49 Hz     | 2300 deg <sup>2</sup>                     |
| 56 Hz     | 1000 deg <sup>2</sup>                     |
| 1024 Hz   | 10 deg <sup>2</sup>                       |